BHOJPURI ASR: HEAD-TO-HEAD MODEL COMPARISON REPORT

Comparative Evaluation of Fine-Tuned Whisper-Small vs. Vakyansh Wav2Vec 2.0 across Datasets, Training Methods, Checkpoints & Accuracy

1. Executive Summary & Benchmark Scorecard

This study presents an empirical head-to-head evaluation between two major state-of-the-art architectures for Bhojpuri speech recognition: a fine-tuned **OpenAI Whisper-Small** (Seq2Seq Transformer) and the open-source **Vakyansh Bhojpuri** (Wav2Vec 2.0 CTC acoustic model). Both models were evaluated on the exact same benchmark split of **1,054 Bhojpuri audio samples** (data/merged_bhojpuri/eval). **Whisper-Small achieved superior linguistic accuracy (40.58% WER)**, while **Vakyansh demonstrated ~5x faster inference throughput**.

Evaluation Metric	Fine-Tuned Whisper-Small	Vakyansh Wav2Vec 2.0	Winner / Core Trade-Off
Best Overall WER (Accuracy)	**40.58% WER (at Checkpoint-14900)**	108.22% WER	■ **Whisper-Small (+67.6% lower error)**
Studio Speech WER (IISc SYSPIN)	**~36.8% WER** (High accuracy)	102.96% WER	■ **Whisper-Small**
Field / Mobile Speech WER (AI4Bharat)	**~44.2% WER** (Noise tolerant)	114.88% WER	■ **Whisper-Small (Context robustness)**
Inference Speed (Throughput)	~4–6 audio files / second	■ **~25 audio files / second**	■ **Vakyansh (~5x faster CTC)**
GPU VRAM Footprint	~2.8 GB VRAM (FP16)	~1.2 GB VRAM (FP16)	■ **Vakyansh (Lighter resource demand)**
Model Parameters & Architecture	240 Million Params (Seq2Seq Transformer)	95 Million Params (Wav2Vec2 + CTC)	Whisper: Autoregressive LM; Vakyansh: CTC

2. Dataset Analysis: Training & Evaluation Corpora

A critical reason for performance differences lies in the volume, acoustic environment, and diversity of the datasets used for training and evaluation:

Category	Whisper-Small Training Data	Vakyansh Training Data	Shared Benchmark Eval Set
Dataset Source	ai4bharat/Rural_Women_Bhojpuri	ULCA / IISc RESPINS & SYSPIN	data/merged_bhojpuri/eval
Total Volume / Hours	~100,000+ Utterances (~150+ Hours)	~60 Hours (Male & Merged)	1,054 Audio Files (~4.8 GB Arrow cache)
Speaker Demographics	Native rural women across rural Bihar & UP districts	Studio voice artists + field contributors (18+ artists)	Diverse mix: 610 Studio + 444 Crowdsourced mobile
Acoustic Conditions	Village environment, natural ambient noise, phone audio	Clean studio recordings + controlled mobile samples	6 Topic Domains (Politics, Health, Food, Finance, etc.)
Target Script	Devanagari script with standard conversational tokens	Pure Devanagari character-level set (65 tokens)	Normalized Devanagari reference ground truth

3. Fine-Tuning Methodology & Architectural Comparison

The two models adopt fundamentally different paradigms for Automatic Speech Recognition:

Dimension	Fine-Tuned Whisper-Small Approach	Vakyansh Wav2Vec 2.0 Approach
Underlying Architecture	**Encoder-Decoder Transformer** (Seq2Seq). Audio spectrograms are mapped into an encoder; an autoregressive decoder generates text token by token.	**CTC Acoustic Model** (Encoder-only). Raw 16 kHz waveform is processed via 1D CNNs + Transformer layers; a linear head outputs character logits.
Loss Function & Training	Cross-Entropy Loss with Teacher Forcing via Hugging Face Seq2SeqTrainer. AdamW optimizer, warmup, cosine decay.	Connectionist Temporal Classification (CTC) Loss via Fairseq. Learns acoustic alignments without explicit temporal alignment labels.
Language Modeling	**Built-in Decoder LM**: Inherently understands Bhojpuri grammar, sentence structure, and context, correcting homophones automatically.	**Acoustic Only (No LM)**: Emits purely phonetic character sequences; requires external KenLM n-gram model to resolve grammar.

Input Representation	80-channel log-magnitude Mel-spectrogram calculated over 25ms windows.	Raw 1D time-domain audio waveform sampled at 16,000 Hz.
Inference Nature	Autoregressive (beam search / greedy). Slower per token but context-aware.	Non-autoregressive (single forward pass argmax). Extremely fast parallel decoding.

4. Checkpoints, Training Progression & History

A breakdown of the checkpoint evolution recorded during model training and benchmark evaluations:

Model & Checkpoint	Step / Epoch	Eval Loss	WER (%)	Status & Notes
Whisper-Small (Initial)	Step 100 (0.01 Ep)	0.6940	71.50%	Baseline adaptation starting point
Whisper-Small (Early)	Step 400 (0.04 Ep)	0.4352	58.88%	Rapid initial phonetic convergence
Whisper-Small (Epoch 1.0)	Step 10,800 (1.03 Ep)	0.2803	42.34%	Passed full dataset once (Saved on disk)
■ Whisper-Small (Best)	**Step 14,900 (1.42 Ep)**	**0.2655**	**40.58%**	**All-Time Best Checkpoint (models/bhojpuri-whisper-small-full)**
Whisper-Small (Current)	Step 15,500 (1.48 Ep)	0.2642	43.80%	Active training checkpoint (~49.3% of 3-epoch goal)
Vakyansh Bhojpuri	bhom_60 (60h)	N/A (CTC)	**108.22%**	Static open-source release checkpoint (102.96% Studio / 114.88% Field)

5. Practical Insights & Deployment Recommendations

- **Why Whisper-Small Excels in Accuracy (40.58% WER):** The multilingual pre-training + integrated Devanagari language model enables Whisper to understand Bhojpuri grammar and sentence context, effectively bridging dialectal noise and phonetic ambiguities.
- **Why Vakyansh Struggles with Word Error Rate (108.22% WER):** Without a language model, CTC predicts purely character-by-character. Minor character omissions split single words into two or merge words (e.g. ■■■■■■■■), which heavily penalizes string-level WER. Furthermore, Vakyansh spells out numbers in full words (*1781* → ■■■■ ■■ ■■■■■■■).
- **When to Use Vakyansh:** Real-time streaming on low-power devices, mobile voice command triggers, and high-throughput audio pre-screening where speed (>25 audios/sec) and low memory (1.2 GB) matter most.
- **When to Use Fine-Tuned Whisper:** Full-length voice transcription, document dictation, conversational AI bots, and subtitles where word accuracy is paramount.
- **Recommended Path Forward:** Resume Whisper-Small fine-tuning from Checkpoint-15500 to reach 3.0 Full Epochs (driving WER toward ~30-35%), while exploring a hybrid pipeline that pairs Vakyansh for instant wake-word detection with Whisper for full utterance transcription.